

Image and Graphics

Unit-3

Image

- An image is a spatial representation of an object, a two dimensional or three-dimensional scenes or another image.
- Abstractly, an image is a continuous function defining a rectangular region of a plane.
- Examples:
 - Recorded image- photographic, analog video signal or in digital format
 - Computer vision- video image, digital image or picture
 - Computer graphics- digital image
 - Multimedia- deals about all above format

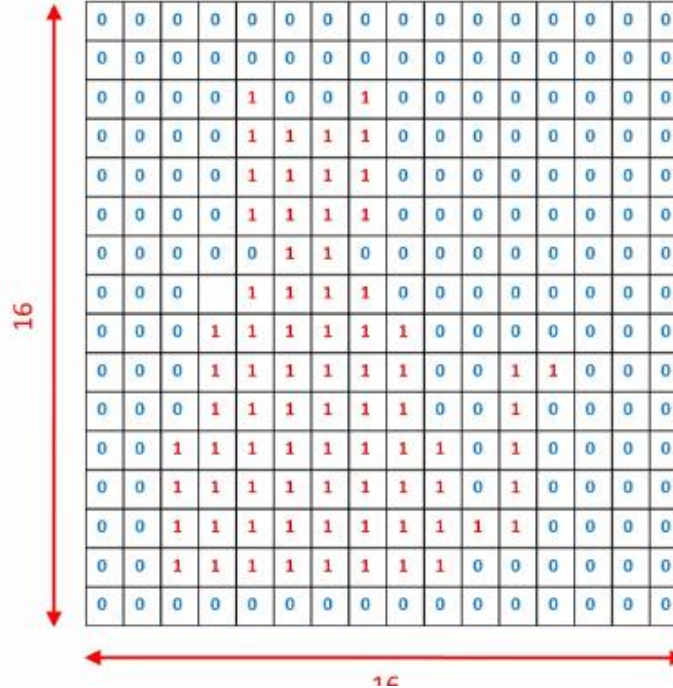
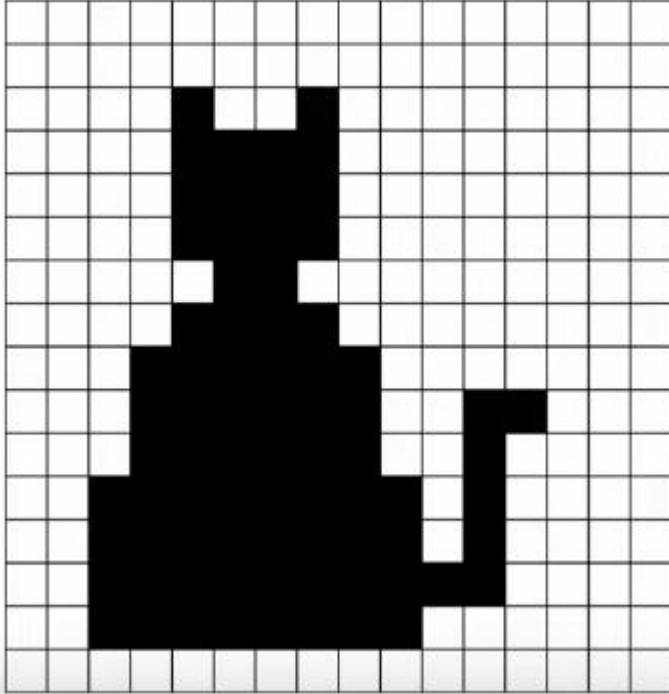
Image vs Graphics

Graphics	Image
Graphics are created normally in graphics application and internally represented as an assemblage of objects such as line, curves or circle.	Image can be from the real world or virtual. Image can be captured from analog signal via scanner or video camera whereas different software tools can used to create virtual too.
The representation is aware of semantics	They ignore the semantic contents
Objects graphics can be individually deleted, added, moved or modified later	Object based editing is not possible, but image editing tools exist for enhancing and retouching bitmap images.

Image vs Graphics

- The smallest addressable image element is called a pixel
- The array and thus the set of pixels, is called a bitmap (Images).
- The drawback of bitmaps is that they need much more storage capacity than graphics.
- Their advantage is that no processing is necessary before displaying them, unlike graphics where abstract definition must be processed first to produce a bitmap.

Bitmap



$16 * 16 = 256$ pixels

Each pixel requires 1 bit

256 bits altogether

$256/8 = 32$ bytes

Bitmap

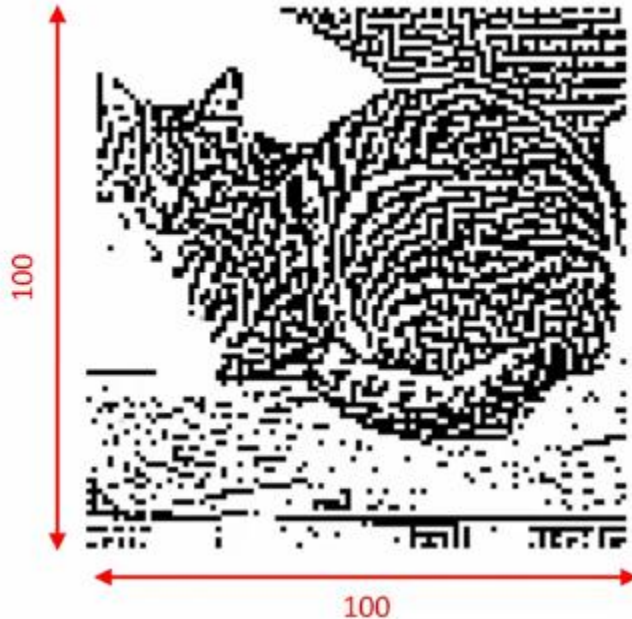
- This was what very early computer could manage
- 1 bit per pixel means monochrome image

Bitmap Resolution

- Smaller pixel fit more in screen. This means having higher density of pixel.
- Image resolution is a measure of pixel density given in dots per inch (dpi)
- If an screen has a resolution of 300 DPI, it means there are 300 pixels (or dots) packed into each inch of the image or display.
- visual representation of how pixel size can vary between different displays

Display Type	Resolution	Pixel Size	Pixel Density (PPI)
Smartphone Screen	1920 × 1080	Smaller	Higher
Computer Monitor	1920 × 1080	Larger	Lower
LED Billboard	1920 × 1080	Much Larger	Much Lower

Bitmap Resolution



$100 * 100 = 10,000$ pixels

Each pixel requires 1 bit

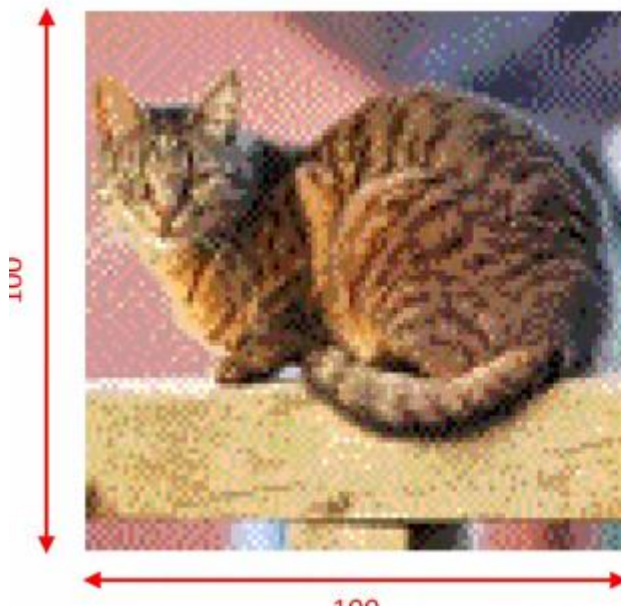
10,000 bits altogether

$10,000/8 = 1,250$ bytes

$1,250/1,024 = 1.22$ KB

Bitmaps in Colour

- 8 bit used to describe the colour of each pixel i.e $2^8 = 256$ colours



$100 * 100 = 10,000$ pixels

Each pixel requires 8 bits

80,000 bits altogether

$80,000 / 8 = 10,000$ bytes

$10,000 / 1,024 = 9.77$ KB

Bitmaps in Colour

- To display the image in colour, we require more bit per pixel.
- For example, here it is using 8 bit per pixel

11001101	11001101	11001101	11001101	11001101
11001101	11001101	11001010	11001010	11001010
11001101	11001101	11001010	11001010	11001010
00101101	00101101	11001010	11001010	11001010
00101101	00101101	00101101	00101101	00101101

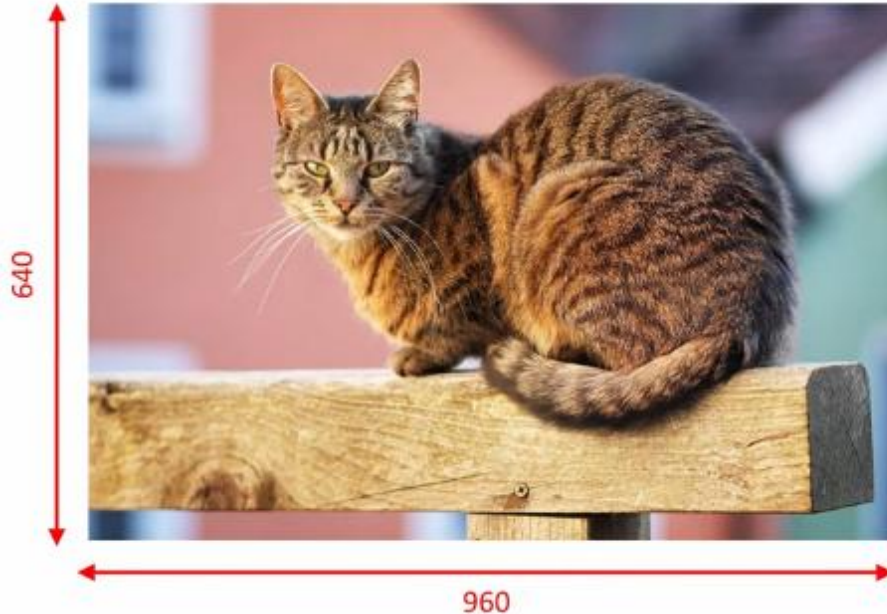
Bitmaps in Colour

- 24 bits used to describe the colour of each pixel i.e $2^{24} = 16,777,216$ colours
- Each byte is used to represent the intensity of a primary colour: Red, Green and Blue. Each colour have 256 different level.

RED	GREEN	BLUE	Colour
255	0	0	Red
0	255	0	Green
0	0	255	Blue
255	255	0	Yellow
255	0	255	Magenta
0	255	255	Cyan
127	127	127	Light gray
255	255	255	White
0	0	0	Black

Bitmaps in Colour

- Calculate the size of image with 24 bit used to describe the colour in each pixel?



Bitmaps in Colour

- Hence, the depth of an image is the number of bits used to represent each pair.



$$960 * 640 = 614,400 \text{ pixels}$$

Each pixel requires 3 bytes

$$614,400 * 3 = 1,843,200 \text{ bytes}$$

$$1,843,200 / 1024 = 1,800 \text{ KB}$$

$$1800 / 1024 = 1.76 \text{ MB}$$

Image Format

- There are different kinds of image formats in the literature. We shall consider only distinguishable two, i.e **Image Capturing format** and **Image Storage Format**.
- That is the format in which the image is created during the digitizing process and the format in which images are stored (and often transmitted).

Image Capturing Format

- The image format is specified by two main parameters: spatial resolution, indicated in pixel and color encoding, which is specified by bits per pixel.
- The value of both parameter depends on the hardware and software used to input and output images

Image Storage Format

- The image is represented in a two-dimensional matrix, in which each value corresponds to the data associated with one image pixel.
- In bitmaps, these values are binary numbers.
- Image file format include:
 - GIF (Graphic Interchange Format)
 - X11 bitmap
 - Postscript
 - JPEG (Joint Picture Expert Group)
 - TIFF (Tagged Image File Format) etc.

Graphics Format

Graphic image formats are specified through graphics primitives and their attributes because they are not specified by the pixels.

- Graphic primitive – line, rectangle, circle, ellipses, specification 2D and 3D objects.
- Graphic attribute – line style, line width, color.
- This higher level has to be mapped to a lower level at one point during image processing, for example, when Computer representing an image.

Graphics Format

- Having the primitives on a higher level is an advantage, because it reduces the data volume that has to be stored for each image and allows simpler image manipulation.
- Having the primitives on a higher level is an advantage, because it reduces the data volume that has to be stored for each image and allows simpler image manipulation.
- Some graphics packages, for example, the SRGP (Simple Raster Graphics Package), include this type of conversion.
- Similarly, PHIGS (Programmer's Hierarchical Interactive Graphics System) and GKS (Graphical Kernel System)

Storage Format

Format	Extension
Microsoft Windows DIB	.bmp .dib .rle
Microsoft Palette	.pal
Autocad format 2D	.dxf
JPEG	.jpg
Windows Meta file	.wmf
Portable network graphic	.png
Compuserve gif	.gif
Apple Macintosh	.pict .pic .pct

Computer Image Processing

- **Computer graphics concern the pictorial synthesis of real or imaginary objects from their computer-based models.**
- **But Image Processing treats in converse process i.e. analysis of scenes or reconstruction of models from pictures of 2D and 3D objects.**
- **Computer image processing comprises of image synthesis (generation) and image analysis (recognition).**

Image Synthesis

- Image synthesis deals with the generation of images of real or imaginary objects. It is an integral part of all computer user interfaces and is indispensable for visualizing 2D, 3D and higher dimensional objects.

Source Pose



Source Image



Target Pose



Synthesize



Target Image



Image Synthesis

- Some of the applications of image synthesis (areas which use image synthesis) are:

1) User Interface

- Point-and-click facility
- Menu-driven

2) Office automation and electronic publishing

- Desktop publishing
- Electronic publishing
- Hypermedia Systems

Image Synthesis

3) Simulation and Animation for Scientific Visualization and Entertainment

- Simulation of real time systems
- Visualization of time-varying behavior of systems
- Abstract representation of complex mathematical expressions
- Models for fluid flow, chemical reaction etc.
- Cartoons
- Flying logos and more exciting visual for movies

Image Synthesis can be dynamic. Similarly, interactive graphics systems are used for image synthesis.

Dynamics in Graphics

- The use of graphics is not limited to static images. Images can also vary dynamically.
- For example, a user can control an animation by adjusting the speed or changing the visible part of a scene or a detail.
- This means that dynamics are an integral part of (dynamic) graphics.
- Most modern interactive graphics technologies include hardware and software allowing users to control and adapt the dynamics of a presentation:

Dynamics in Graphics

Dynamics can be of two kinds:

- **Motion Dynamics**
 - **Movement dynamics means that objects can be moved or activated relative to a static observer's viewpoint.**
 - **Objects may also be static while their environment moves.**
 - **A typical example is flight simulator**

- **Update Dynamics**
 - **Update Dynamics is the actual change of the shape, color, or other properties of the objects being viewed.**

Framework of interactive graphics system

- Graphical images are generated using interactive graphics systems (e.g., SRGP).
- The high level conceptual framework of almost any interactive graphics system consists of three software components:
 - An application model
 - An application program
 - A graphics system

An application model

- It represents the data or objects to be pictured on the screen.
- It typically stores descriptions of **primitives** that define the shape of components of the object, object attributes and connectivity relationships that describe how the components fit together.
- It is stored in an **application database**.
- The model is application-specific, and is created **independently** of any display system

An application program

- **It converts the description of the primitives from the application model into the procedure calls or commands the graphics system uses to create an image.**
- **This conversion involves two steps:**
 - **Extraction of the portions to be viewed (from the application database) using some selection or query system**
 - **Conversion of the extracted geometry into proper format**
- **Hence, the application program produces views by sending to the graphics system, a series of graphics output commands.**

A graphics System

- The graphics system is an intermediary between the application program and the display hardware.
- It is responsible for actually producing the picture. It performs two kinds of transformations:
 - Output transformations (from objects in the application model to a view of the model)
 - Input transformations (from user actions to application program inputs that cause the application to make changes in the model and/or/picture)

Graphics Hardware

- This is the hardware that receives input from interaction devices and outputs images to display devices.
- Below are given the input and output to the graphics hardware.
 - Graphics Hardware - Input
 - Graphics Hardware - Output



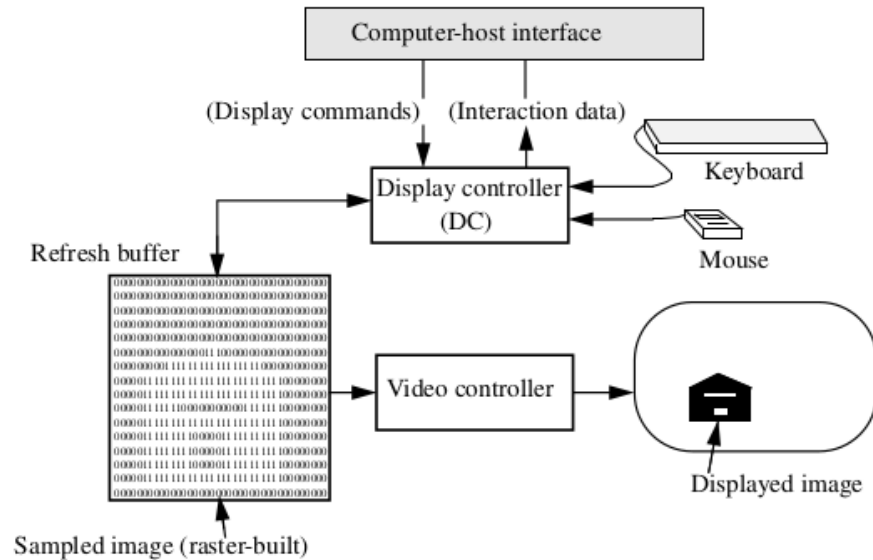
Graphics Hardware – Input

- Input to the graphics hardware is provided through the input devices such as the mouse, the data tablet, the touch panel, etc.
- Other devices include trackballs, spaceballs and the data glove. Even audio communication is used for providing input to graphics hardware.



Graphics Hardware – Output

- Modern visual output technologies use raster display devices that display primitives in special buffers called refresh buffers in the form of pixel components.



Architecture of a raster display device.

Graphics Hardware – Output

- Some raster display devices have a display controller in hardware that receives and interprets output command sequences.
- In low-end systems, for example, personal computers the display controller is implemented in software and exists as a component of a graphics library.
- In this case, the refresh buffer is part of the CPU memory.
- The video controller can read this memory to output images.
- The complete display of a raster display device is formed by the raster, consisting of horizontal raster lines that are each composed of pixels.
- The video controller samples the entire screen line by line.

Dithering

- Dithering is a process used in output devices (monitor screens and some printers) to produce shades of gray or tints of colors.
- The idea is to visually produce an effect on the image that is as close to the original image as possible but using less color.
- Dithering uses the eye's capability for spatial integration, which means that if we view a very small area from a sufficiently large viewing distance, our eyes average fine detail within the small area and record only the overall intensity of the area.



Dithering

- Dithering use two techniques:
 - Clustered dot ordered dithering (halftoning)
 - Dispersed-dot ordered dithering

Original



Halftoned



Clustered-dot ordered dithering

- Four adjacent pixels are used for presenting the desired intensity of a single color.
- By illuminating none, any one, two, three or all four of the adjacent pixels, different intensities of the color are displayed.
- The dither matrix can be used to represent the patterns used.
- This technique is used in devices that are not able to display individual dots (e.g., laser printers).

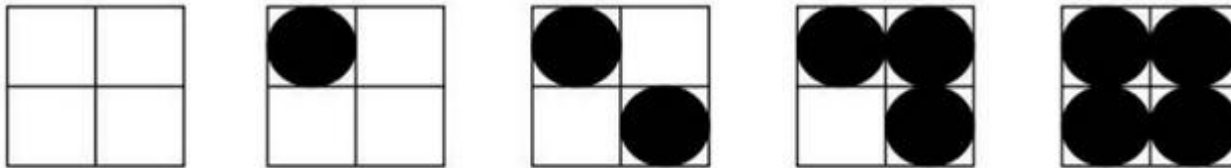


Fig.: Five intensity levels approximated with four 2 X 2 dither patterns.

Dispersed-dot ordered dithering

- This technique is used for devices such as CRT displays that can display individual dots. Using the 2 X 2 dither pattern, a color CRT display can present 125 colors.
- Each pattern can display five intensities for each color (Red, Green or Blue), by using the halftone patterns as in clustered-dot ordered dithering. Hence total number of resultant intensities = $5 \times 5 \times 5 = 125$ color combinations.



Dithering

- Thus dithering is a process of creating the illusion of new colors and shades by varying the pattern of dots. Dithering is usually called halftoning, and shades of gray are called halftones.

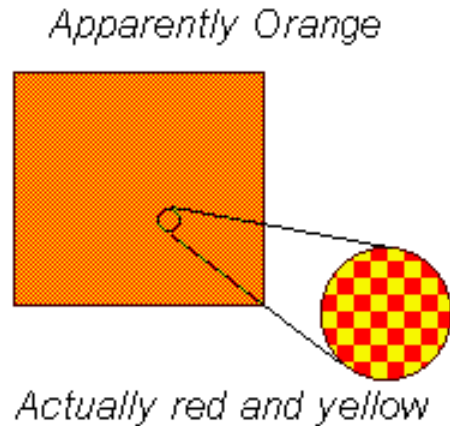


Image Analysis

- **Image analysis is concerned with techniques for extracting descriptions of objects from images.**
- **Image analysis techniques include computation of perceived brightness and color, partial or complete recovery of three-dimensional data in the scene, location of discontinuities corresponding to objects in the scene and characterization of the properties of uniform regions in the image.**
- **It may deal with calculating and comparing the intensity, hue, saturation of the image, the centroid, and identification of noise, detection of patterns or recognition of the image itself**

Image Analysis

Image analysis is used in areas such as

- **Aerial surveillance photographs**
- **Slow scan images of the moon or of planets gathered from space probes**
- **Television images taken from an industrial robot's visual sensor**
- **X-ray images and computerized axial tomography (CAT) scans, etc.**

Image Recognition

- To fully recognize an object in an image means knowing that there is an agreement between the sensory projections and the observed image.
- How the object appears in the image is specified by the spatial configuration of the pixel values. Thus, agreement between the observed spatial configuration and the expected sensory projections is required.

Image Recognition

This, in turn, requires the following capabilities.

- **Infer explicitly or implicitly an object's position and orientation from the spatial configuration.**

This requires the capability to infer which pixels are parts of the object. Also, object features such as special markings, curves, lines, etc. have to be distinguished.

- **Confirm that the inference is correct.**

This depends on matching the distinguishing image features with corresponding object features.

Image Recognition

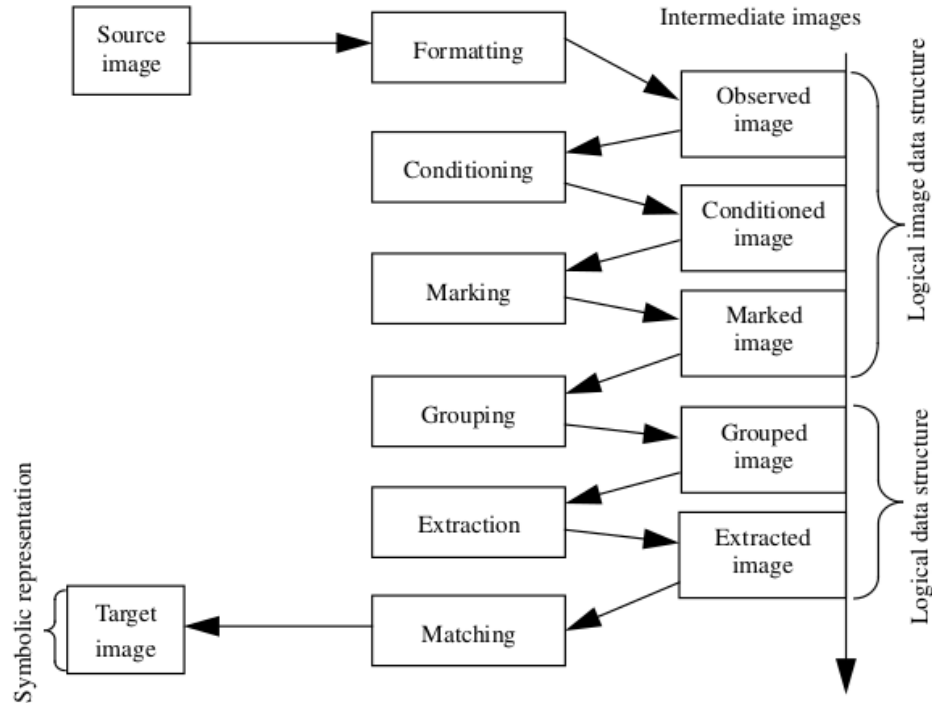


Figure 4-13 Steps involved in image recognition.

Image Recognition Steps

1. Formatting

- It refers to capturing an image from a camera and bringing it into a digital form, i.e., generating a digital representation of an image in the form of pixels.

1. Conditioning

- Prepare and preprocessing input data to make suitable for recognition
- Task may include noise reduction, image enhancement normalization or feature extraction

Image Recognition Steps

3. Labeling

- Assign semantic labels or categories to objects or pattern
- Annotate images with class labels corresponding to the recognized features.

3. Grouping

- Organize or group similar objects or patterns
- Use clustering algorithms to group data points or pixel into coherent regions

3. Extracting

- Extract relevant features or descriptions from group objects
- Techniques may include edge detection texture, analysis or shape analysis

Image Recognition Steps

5. Matching

- Compare extracted feature with reference patterns on templates
- Use similarity measures, template matching or machine learning classifiers for comparison

Image Transmission

- Image transmission takes into account transmission of digital images through computer networks.
- There are several requirements on the networks when images are transmitted
 - The network must accommodate bursty transport data, because image transmission is bursty (due to large image sizes).
 - Transport should be reliable.
 - Time-dependence is not dominant as in audio/video transmission.

Image Transmission

- **Image size depends on the image representation format used for transmission. There are several possibilities:**
 - **Raw Image Data Transmission**
 - **Compressed Image Data Transmission**
 - **Symbolic Image Data Transmission**

Raw Image Transmission

- The image is generated through as video digitizer and transmitted in its digital format. The image size is calculated as:

$$\text{Size} = \text{Spatial_Resolution} \times \text{Pixel_Quantization}$$

- For example, the transmission of an image with a resolution of 640 X 480 pixels and pixel quantization of 8 bits per pixel requires $640 \times 480 \times 8 = 2457600$ bits (=307,200 bytes).

Compressed Image Data Transmission

- The image is generated through a video digitizer and compressed before transmission.
- Methods such as JPEG or MPEG are used for compression.
- The reduction of image size depends on the compression method and compression rate.

Symbolic Image Data Transmission

- The image is represented through symbolic data representation as image primitives (e.g., 2D or 3D geometric representation), attributes and other control information.
- This image representation method is used in computer graphics.
- Image size is equal to the structure size, which carries the transmitted symbolic information of the image.